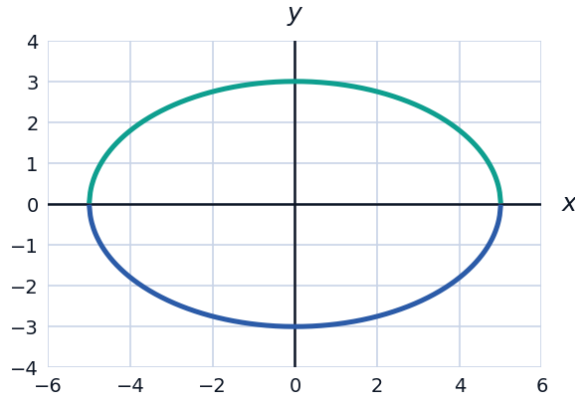


Ellipses



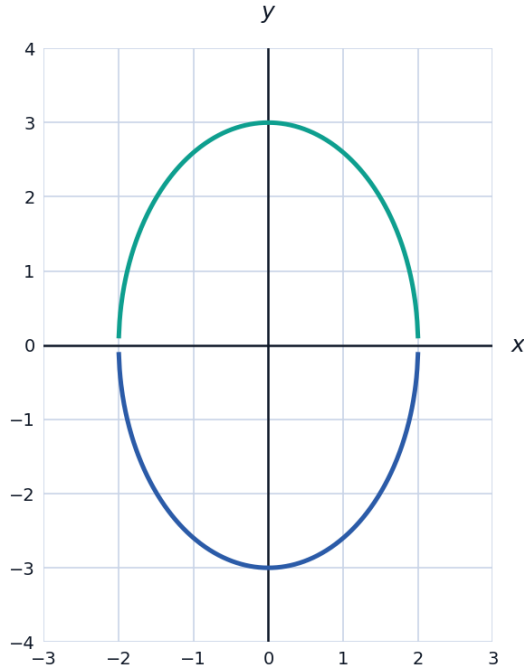
Standard form

- 1 For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find:



- a** the vertices
 - b** the foci
 - c** the endpoints of the minor axis
 - d** the eccentricity
- 2 Write the standard-form equation of the ellipse with foci $(0, \pm 3)$ and vertices $(0, \pm 5)$.

- 3 Consider the ellipse $9x^2 + 4y^2 = 36$.



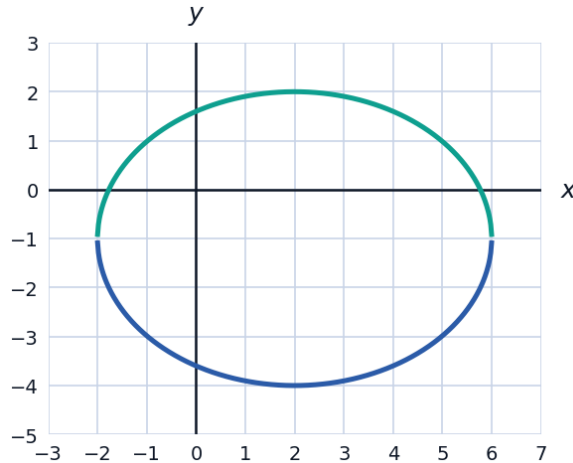
- a Write the equation in standard form.
- b State the lengths of the major and minor axes.
- c Find the coordinates of the foci.

Applications

- 4 A whispering gallery has an elliptical ceiling 40 m long and 24 m high at the center (so $a = 20$ and $b = 12$, with the major axis horizontal). Two people stand at the foci. How far apart are they, to one decimal place?
- 5 An ellipse centered at the origin has a horizontal major axis of length 12 and passes through the point $(3, 2\sqrt{3})$. Find its equation.

Ellipses centered away from the origin

- 6 For the ellipse $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$, find:



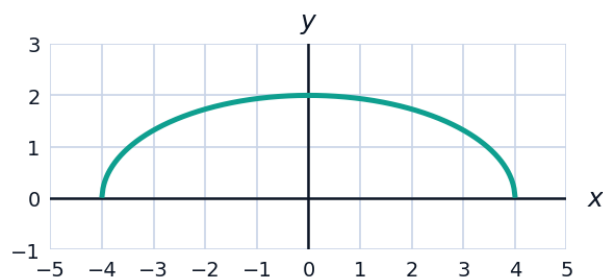
- a** the center **b** the vertices **c** the foci
- 7 Complete the square to write $4x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, and state the center.

Eccentricity and orbital applications

- 8 Compute the eccentricity of each ellipse, and state which is more circular (closer to $e = 0$):
- a** $\frac{x^2}{100} + \frac{y^2}{36} = 1$ **b** $\frac{x^2}{25} + \frac{y^2}{24} = 1$
- 9 A planet's elliptical orbit has semi-major axis $a = 150$ million km and eccentricity $e = 0.02$. Find the distance from the center to a focus, c , and hence the closest approach distance to the sun ($a - c$), to two decimal places (in millions of km).

Graphing an ellipse

- 10 The upper half of the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$ is shown (as $y = 2\sqrt{1 - x^2/16}$). Use it to estimate the y -value at $x = 2$, then verify algebraically.

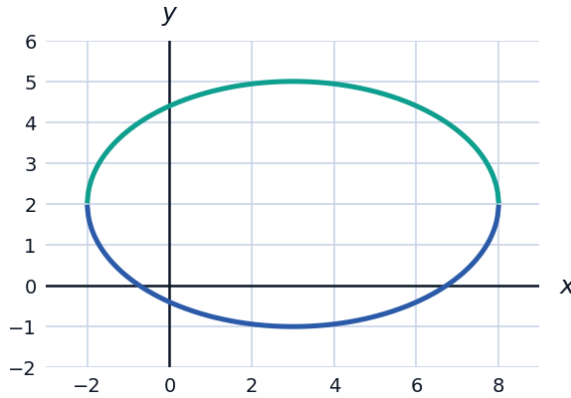


The ellipse's defining property

- 11 State the geometric (sum-of-distances) definition of an ellipse, and use it to verify that the point $(0, 3)$ lies on the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with foci $(\pm\sqrt{7}, 0)$.
-

Writing equations from a description

- 12 Write the standard-form equation of an ellipse centered at $(3, 2)$ with a horizontal major axis of length 10 and minor axis of length 6.



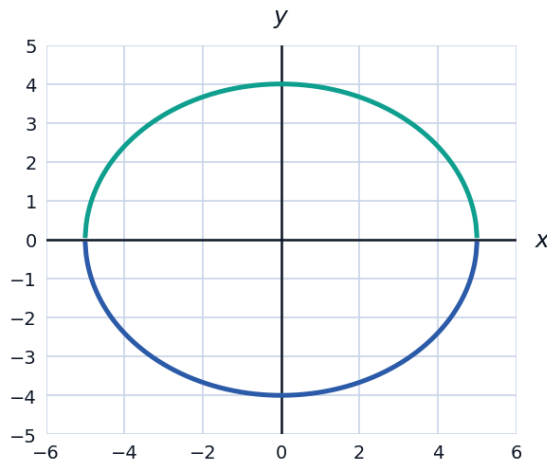
- 13 Write the standard-form equation of an ellipse with vertical major axis, center $(-1, 4)$, $a = 7$, and $b = 2$.
-

Systems with ellipses

- 14 Find the points of intersection of $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the line $y = x$.
-

Area of an ellipse

- 15 State the formula for the area of an ellipse with semi-axes a and b , and use it to find the area of $\frac{x^2}{25} + \frac{y^2}{16} = 1$.



- 16 An elliptical track for a running field has a semi-major axis of 60 m and semi-minor axis of 40 m. Find the exact area enclosed by the track, and give a decimal approximation to the nearest square metre.

Comparing ellipses of the same eccentricity

- 17 Two ellipses both have eccentricity $e = 0.6$: one has $a = 10$, the other has $a = 20$. Find b for each, and explain why they are the same shape but different sizes (similar ellipses).